Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Valentine Dairy

Physical address of dairy:

20717 Bloss AVE	Hilmar	Merced	95324
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1932

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X170-X034-XXXX

B. OPERATORS

Clauss-Jorristma, Kimberly

Operator name: Clauss-Jorristma, Kimberly

Telephone no.: (209) 667-4582 (Landline) (Cellular)

21670 Bloss AVE	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Clauss-Jorristma, Kimberly

Legal owner name: Clauss-Jorristma, Kimberly

Telephone no.: (209) 667-4582 (Landline) (Cellular)

21670 Bloss AVE	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	412	0	0
Number under roof	0	0	394	0	0	0
Maximum number	0	0	394	412	0	0
Average number	0	0	394	412	0	0
Avg live weight (lbs)	0	0	1,300	900		

Predominant milk cow breed: Other

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 8,678.51 *tons per reporting period*

Total nitrogen from manure: 76,489.40 *lbs per reporting period*

After ammonia losses (30% loss applied): 53,542.58 *lbs per reporting period*

Total phosphorus from manure: 12,944.36 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 19,031,500 *gallons*

Total nitrogen generated: 47,091.22 *lbs*

Total phosphorus generated: 8,546.21 *lbs*

Total potassium generated: 67,919.41 *lbs*

Total salt generated: 410,021.02 *lbs*

	23,306,500	*gallons applied*
+	0	*gallons exported*
-	4,275,000	*gallons imported*
=	19,031,500	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

Date	Material type / Description	Quantity	N (mg/L)	P (mg/L)	K (mg/L)	Salt (mg/L)	TDS (mg/L)
04/21/2023	Process wastewater *WW-CDF*	1,935,000.00 *gal*	342.00	56.90	401.00		2,920
06/19/2023	Process wastewater *WW-CDF*	1,080,000.00 *gal*	342.00	56.90	401.00		2,920
07/04/2023	Process wastewater *WW-CDF*	1,260,000.00 *gal*	450.00	50.00	473.00		3,330

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	13,336.39	1,957.35	15,062.68	108,481.66
Total imports for all materials	13,336.39	1,957.35	15,062.68	108,481.66

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
June	53	53	2	both	X017-X100-X001-XXXX X017-X100-X002-XXXX
Rooster	48	48	3	process wastewater	X015-X020-X033-XXXX X015-X020-X034-XXXX X045-X170-X034-XXXX X045-X170-X058-XXXX
Tell	22	22	3	process wastewater	X015-X020-X034-XXXX X017-X100-X005-XXXX X045-X200-X050-XXXX X045-X200-X072-XXXX
Totals for areas that were used for application	123	123	8		
Totals for areas that were not used for application					
Land application area totals	123	123	8		

B. CROPS AND HARVESTS

June

Field name: June

11/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 53 Plant date: 11/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	398.62 *ton*	As-is		59.2	10,900.00	1,900.00	16,500.00		9.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	7.52	163.96	28.58	248.20	583.04

June

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 53 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	1,778.94 *ton*	As-is		68.6	4,300.00	900.00	5,000.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	33.56	288.66	60.42	335.65	1,285.80

Rooster

Field name: Rooster

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 48 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/16/2023	485.67 *ton*	As-is		70.0	10,200.00	1,300.00	11,900.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.12	206.41	26.31	240.81	582.80

04/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 48 Plant date: 04/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/17/2023	1,698.21 *ton*	As-is		68.5	4,500.00	1,000.00	5,400.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	35.38	318.41	70.76	382.10	1,315.05

Rooster

08/19/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 48 Plant date: 08/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	239.70 *ton*	As-is		73.8	7,400.00	900.00	8,500.00		14.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	4.99	73.91	8.99	84.89	366.34

Tell

Field name: Tell

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 22 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/16/2023	224.24 *ton*	As-is		70.0	8,600.00	1,300.00	11,100.00		9.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.19	175.31	26.50	226.28	556.52

04/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 22 Plant date: 04/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/17/2023	743.64 *ton*	As-is		67.8	4,300.00	1,100.00	5,500.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	33.80	290.70	74.36	371.82	1,393.18

Tell

08/19/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 22 Plant date: 08/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	108.10 *ton*	As-is		73.7	6,200.00	1,400.00	11,700.00		18.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	4.91	60.93	13.76	114.98	483.32

NUTRIENT BUDGET

A. LAND APPLICATIONS

June - 11/29/2022: Oats, silage-soft dough

Field name: June

Crop: Oats, silage-soft dough — Plant date: 11/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	158.28	22.49	30.27	4,878.07	600.00 *ton*
Application event totals		158.28	22.49	30.27	4,878.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	424.75	12,547,150.00 *gal*
Application event totals		0.00	0.00	0.00	424.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.94	7.77	36.33	295.54	861,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	281.33	8,310,450.00 *gal*
Application event totals		32.94	7.77	36.33	576.86	

June - 05/31/2023: Corn, silage

Field name: June

Crop: Corn, silage — Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

June - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.70	10.55	87.79	455.66	885,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	329.41	9,730,680.00 *gal*
Application event totals		51.70	10.55	87.79	785.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.00	12.85	106.98	555.29	1,078,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	234.38	6,923,655.00 *gal*
Application event totals		63.00	12.85	106.98	789.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	179.31	5,296,901.00 *gal*
Application event totals		0.00	0.00	0.00	179.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	192.66	5,691,240.00 *gal*
Application event totals		0.00	0.00	0.00	192.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	120.42	3,557,108.00 *gal*
Application event totals		0.00	0.00	0.00	120.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	151.09	4,463,110.00 *gal*
Application event totals		0.00	0.00	0.00	151.09	

June - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.44	11.44	116.00	660.53	1,279,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	123.51	3,648,360.00 *gal*
Application event totals		64.44	11.44	116.00	784.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	140.05	4,137,210.00 *gal*
Application event totals		0.00	0.00	0.00	140.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.93	9.93	100.67	573.25	1,110,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	145.92	4,310,450.00 *gal*
Application event totals		55.93	9.93	100.67	719.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.71	10.60	107.47	611.99	1,185,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	104.00	3,072,030.00 *gal*
Application event totals		59.71	10.60	107.47	715.98	

Rooster - 10/28/2022: Oats, silage-soft dough

Field name: Rooster

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Rooster - 10/28/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.36	10.46	48.92	397.95	1,050,000.00 *gal*
Application event totals		44.36	10.46	48.92	397.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.43	11.66	54.51	443.43	1,170,000.00 *gal*
Application event totals		49.43	11.66	54.51	443.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.03	13.45	62.90	511.65	1,350,000.00 *gal*
Application event totals		57.03	13.45	62.90	511.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.46	13.55	63.37	515.44	1,360,000.00 *gal*
Application event totals		57.46	13.55	63.37	515.44	

Rooster - 04/22/2023: Corn, silage

Field name: Rooster

Crop: Corn, silage

Plant date: 04/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	115.05	19.14	134.90	982.31	1,935,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	336.39	8,999,638.00 *gal*
Application event totals		115.05	19.14	134.90	1,318.71	

Rooster - 04/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	16.45	3.36	27.93	144.97	255,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	250.62	6,704,789.00 *gal*
Application event totals		16.45	3.36	27.93	395.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	129.18	3,456,079.00 *gal*
Application event totals		0.00	0.00	0.00	129.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	64.21	10.68	75.29	548.27	1,080,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	167.45	4,479,918.00 *gal*
Application event totals		64.21	10.68	75.29	715.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	142.46	3,811,310.00 *gal*
Application event totals		0.00	0.00	0.00	142.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	98.58	10.95	103.61	729.46	1,260,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	194.58	5,205,649.00 *gal*
Application event totals		98.58	10.95	103.61	924.04	

Rooster - 04/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	194.58	5,205,649.00 *gal*
Application event totals		0.00	0.00	0.00	194.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	145.14	3,883,008.00 *gal*
Application event totals		0.00	0.00	0.00	145.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.06	4.45	45.11	256.89	450,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	131.74	3,524,518.00 *gal*
Application event totals		25.06	4.45	45.11	388.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	109.31	2,924,349.00 *gal*
Application event totals		0.00	0.00	0.00	109.31	

Rooster - 08/19/2023: Sudangrass, silage

Field name: Rooster

Crop: Sudangrass, silage

Plant date: 08/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Rooster - 08/19/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.39	7.70	78.11	444.79	780,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	299.79	8,020,399.00 *gal*
Application event totals		43.39	7.70	78.11	744.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	269.22	7,202,390.00 *gal*
Application event totals		0.00	0.00	0.00	269.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.08	9.78	99.14	564.54	990,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	323.06	8,642,868.00 *gal*
Application event totals		55.08	9.78	99.14	887.60	

Tell - 10/28/2022: Oats, silage-soft dough

Field name: Tell

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.42	0.00	0.00	0.00	330,000.00 *gal*
Application event totals		30.42	0.00	0.00	0.00	

Tell - 10/28/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.18	0.00	0.00	0.00	360,000.00 *gal*
Application event totals		33.18	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.66	0.00	0.00	0.00	810,000.00 *gal*
Application event totals		74.66	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.48	0.00	0.00	0.00	450,000.00 *gal*
Application event totals		41.48	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	13.83	0.00	0.00	0.00	150,000.00 *gal*
Application event totals		13.83	0.00	0.00	0.00	

Tell - 04/21/2023: Corn, silage

Field name: Tell

Crop: Corn, silage

Plant date: 04/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.88	11.20	93.20	483.74	390,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	284.69	3,490,902.00 *gal*
Application event totals		54.88	11.20	93.20	768.44	

Tell - 04/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	158.98	1,949,395.00 *gal*
Application event totals		0.00	0.00	0.00	158.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	75.99	15.51	129.04	669.80	540,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	147.33	1,806,512.00 *gal*
Application event totals		75.99	15.51	129.04	817.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	181.61	2,226,923.00 *gal*
Application event totals		0.00	0.00	0.00	181.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	107.46	1,317,662.00 *gal*
Application event totals		0.00	0.00	0.00	107.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	107.46	1,317,662.00 *gal*
Application event totals		0.00	0.00	0.00	107.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.98	9.05	91.76	522.55	420,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	107.46	1,317,662.00 *gal*
Application event totals		50.98	9.05	91.76	630.01	

Tell - 04/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.06	7.11	72.10	410.57	330,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	178.91	2,193,820.00 *gal*
Application event totals		40.06	7.11	72.10	589.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	107.46	1,317,662.00 *gal*
Application event totals		0.00	0.00	0.00	107.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	95.59	16.97	172.06	979.78	787,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	235.53	2,887,987.00 *gal*
Application event totals		95.59	16.97	172.06	1,215.30	

Tell - 08/19/2023: Sudangrass, silage

Field name: Tell

Crop: Sudangrass, silage

Plant date: 08/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.45	7.00	71.01	404.35	325,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	527.58	6,469,115.00 *gal*
Application event totals		39.45	7.00	71.01	931.93	

Tell - 08/19/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.66	7.22	73.19	416.79	335,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	328.51	4,028,124.00 *gal*
Application event totals		40.66	7.22	73.19	745.30	

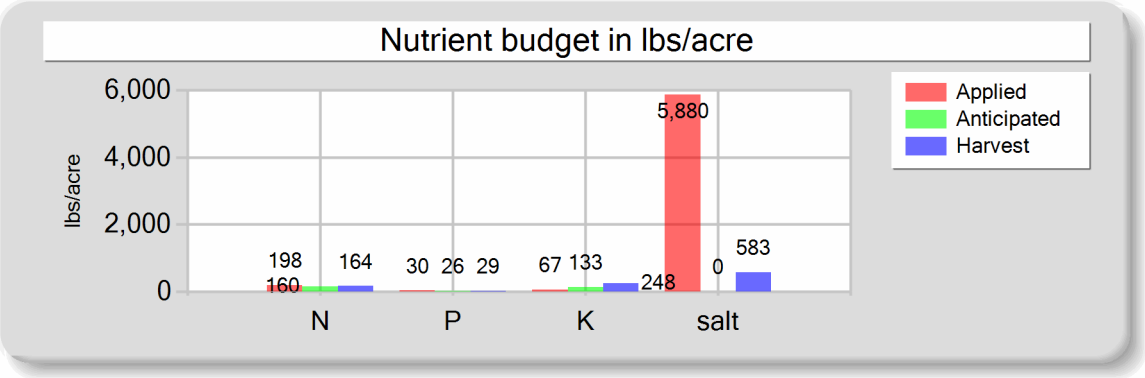
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	338.87	4,155,225.00 *gal*
Application event totals		0.00	0.00	0.00	338.87	

B. NUTRIENT BUDGET

June - 11/29/2022: Oats, silage-soft dough

Field name: June | Crop: Oats, silage-soft dough | Plant date: 11/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	158.28	22.49	30.27	4,878.07
Process wastewater	32.94	7.77	36.33	295.54
Fresh water	0.00	0.00	0.00	706.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.22	30.26	66.60	5,879.68
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	163.96	28.58	248.20	583.04
Nutrient balance	34.26	1.68	-181.59	5,296.64
Applied to removed ratio	1.21	1.06	0.27	10.08

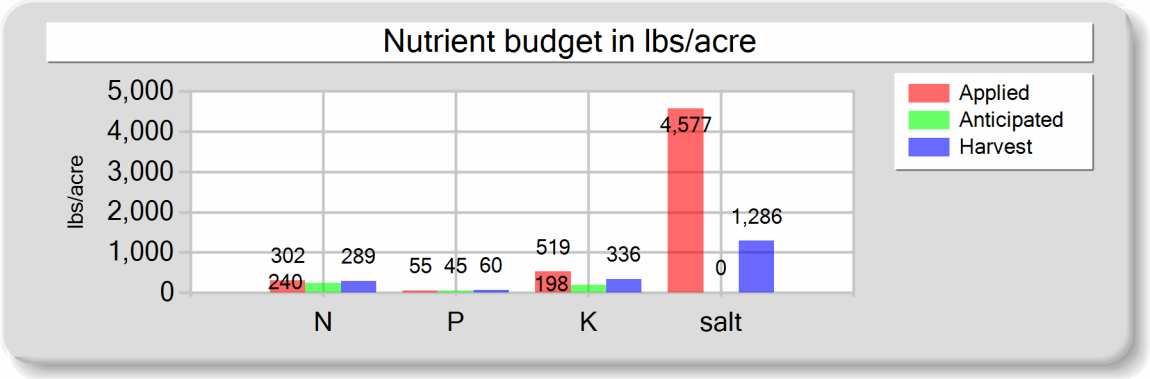
Fresh water applied	
20,857,600.00	*gallons*
768.11	*acre-inches*
14.49	*inches/acre*

Process wastewater applied	
861,000.00	*gallons*
31.71	*acre-inches*
0.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

June - 05/31/2023: Corn, silage

Field name: June | Crop: Corn, silage | Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	294.77	55.37	518.91	2,856.72
Fresh water	0.00	0.00	0.00	1,720.74
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	301.77	55.37	518.91	4,577.46
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	288.66	60.42	335.65	1,285.80
Nutrient balance	13.12	-5.05	183.26	3,291.66
Applied to removed ratio	1.05	0.92	1.55	3.56

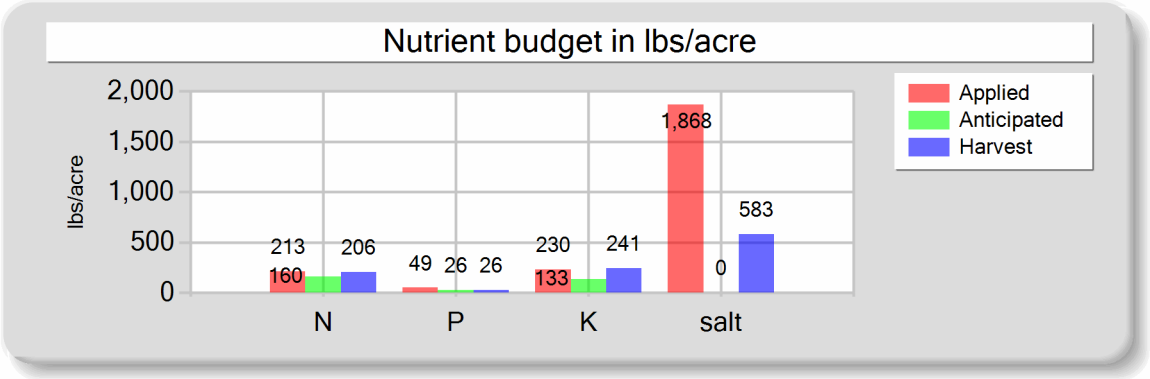
Fresh water applied	
50,830,744.00	*gallons*
1,871.92	*acre-inches*
35.32	*inches/acre*

Process wastewater applied	
5,537,500.00	*gallons*
203.93	*acre-inches*
3.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

Rooster - 10/28/2022: Oats, silage-soft dough

Field name: Rooster | Crop: Oats, silage-soft dough | Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	208.28	49.11	229.70	1,868.48
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	212.94	49.11	229.70	1,868.48
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	206.41	26.31	240.81	582.80
Nutrient balance	6.53	22.80	-11.11	1,285.68
Applied to removed ratio	1.03	1.87	0.95	3.21

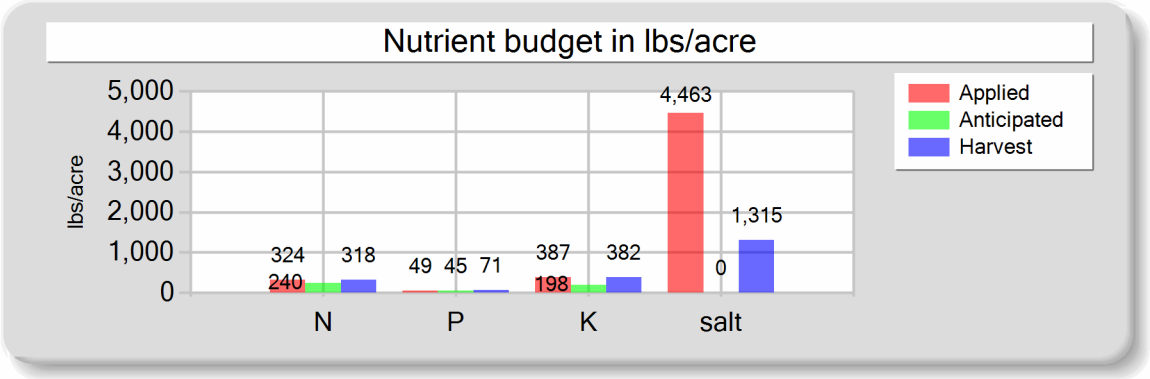
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
4,930,000.00	*gallons*
181.56	*acre-inches*
3.78	*inches/acre*

Total harvests for the crop	
1	*harvests*

Rooster - 04/22/2023: Corn, silage

Field name: Rooster | Crop: Corn, silage | Plant date: 04/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	319.35	48.58	386.85	2,661.90
Fresh water	0.00	0.00	0.00	1,801.46
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	324.02	48.58	386.85	4,463.36
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	318.41	70.76	382.10	1,315.05
Nutrient balance	5.60	-22.18	4.75	3,148.31
Applied to removed ratio	1.02	0.69	1.01	3.39

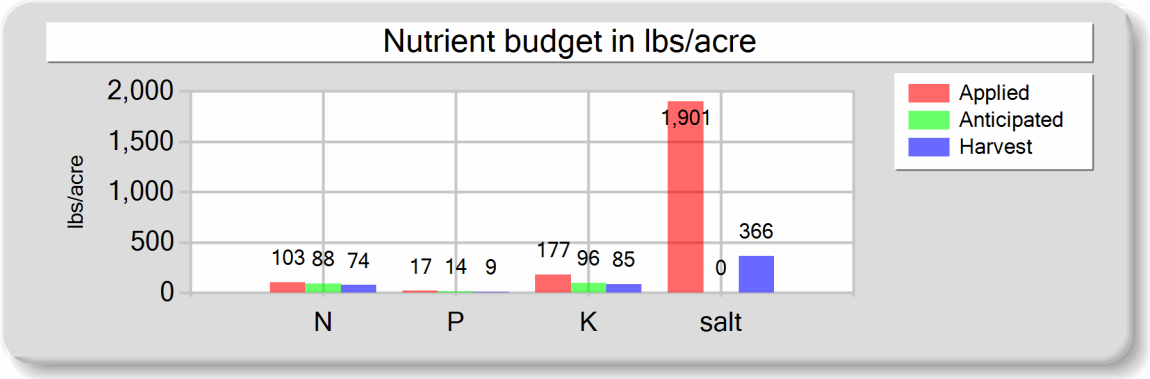
Fresh water applied	
48,194,907.00	*gallons*
1,774.85	*acre-inches*
36.98	*inches/acre*

Process wastewater applied	
4,980,500.00	*gallons*
183.41	*acre-inches*
3.82	*inches/acre*

Total harvests for the crop	
1	*harvests*

Rooster - 08/19/2023: Sudangrass, silage

Field name: Rooster | Crop: Sudangrass, silage | Plant date: 08/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	98.47	17.48	177.25	1,009.33
Fresh water	0.00	0.00	0.00	892.07
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	103.14	17.48	177.25	1,901.39
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	73.91	8.99	84.89	366.34
Nutrient balance	29.23	8.49	92.35	1,535.05
Applied to removed ratio	1.40	1.94	2.09	5.19

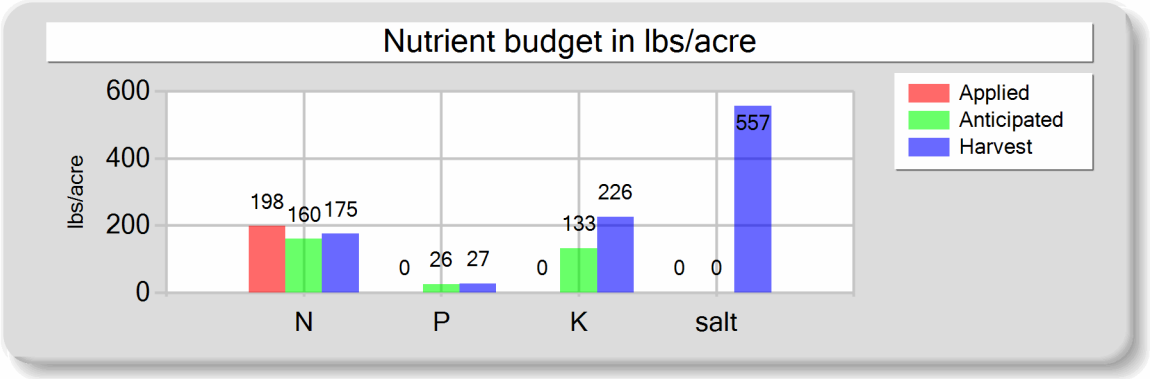
Fresh water applied	
23,865,657.00	*gallons*
878.89	*acre-inches*
18.31	*inches/acre*

Process wastewater applied	
1,770,000.00	*gallons*
65.18	*acre-inches*
1.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

Tell - 10/28/2022: Oats, silage-soft dough

Field name: Tell | Crop: Oats, silage-soft dough | Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	193.57	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	198.23	0.00	0.00	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	175.31	26.50	226.28	556.52
Nutrient balance	22.92	-26.50	-226.28	-556.52
Applied to removed ratio	1.13	0.00	0.00	0.00

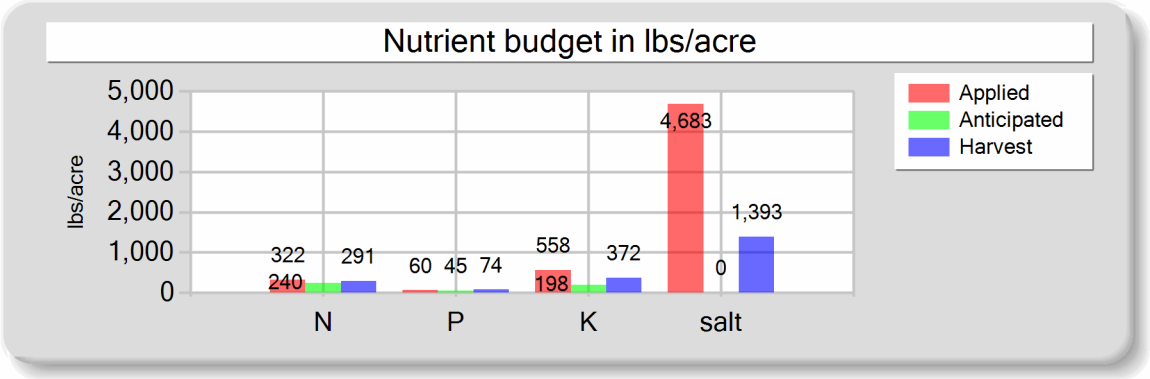
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
2,100,000.00	*gallons*
77.34	*acre-inches*
3.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

Tell - 04/21/2023: Corn, silage

Field name: Tell | Crop: Corn, silage | Plant date: 04/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	317.50	59.83	558.17	3,066.45
Fresh water	0.00	0.00	0.00	1,616.89
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	322.17	59.83	558.17	4,683.34
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	290.70	74.36	371.82	1,393.18
Nutrient balance	31.47	-14.53	186.35	3,290.16
Applied to removed ratio	1.11	0.80	1.50	3.36

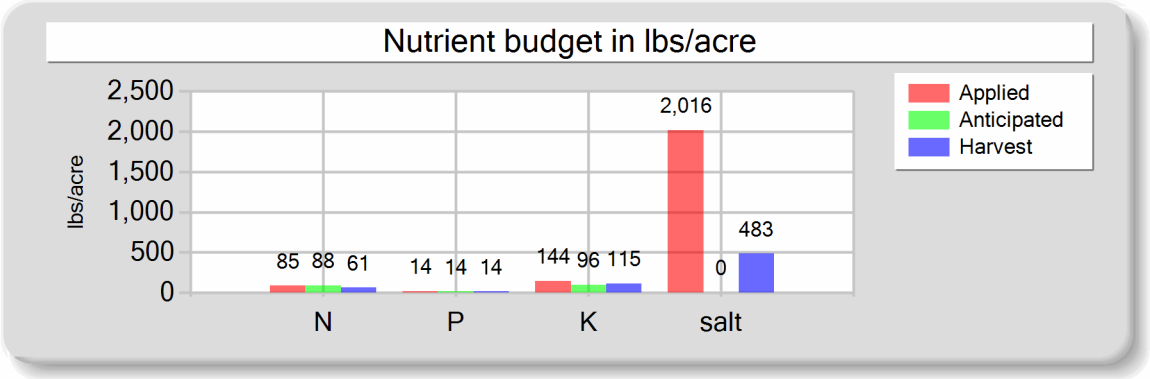
Fresh water applied	
19,826,187.00	*gallons*
730.13	*acre-inches*
33.19	*inches/acre*

Process wastewater applied	
2,467,500.00	*gallons*
90.87	*acre-inches*
4.13	*inches/acre*

Total harvests for the crop	
1	*harvests*

Tell - 08/19/2023: Sudangrass, silage

Field name: Tell | Crop: Sudangrass, silage | Plant date: 08/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	80.11	14.22	144.20	821.15
Fresh water	0.00	0.00	0.00	1,194.96
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	84.78	14.22	144.20	2,016.11
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	60.93	13.76	114.98	483.32
Nutrient balance	23.85	0.46	29.22	1,532.79
Applied to removed ratio	1.39	1.03	1.25	4.17

Fresh water applied	
14,652,464.00	*gallons*
539.60	*acre-inches*
24.53	*inches/acre*

Process wastewater applied	
660,000.00	*gallons*
24.31	*acre-inches*
1.10	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall 2022

Sample and source description: Fall 2022

Sample date: 09/21/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,300.00	2,600.00	3,500.00							56.40
DL	100.00	100.00	100.00							0.01

Spring Corral Solids

Sample and source description: Spring Corral Solids

Sample date: 08/16/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 3.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,800.00	3,200.00	13,500.00							74.20
DL	100.00	100.00	100.00							0.01

Fall Separated Solids

Sample and source description: Fall Separated Solids

Sample date: 12/11/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,800.00	800.00	1,400.00							28.20
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 11/17/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	243.00	136.00	0.00	0.00	57.30	268.00								4,600.00	2,180
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 05/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	371.00	271.00	0.00	0.00	75.70	630.00								6,540.00	3,270
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 08/17/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	320.00	257.00	0.00	0.00	56.80	576.00								6,440.00	3,280
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 12/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	447.00	298.00	0.00	0.00	53.80	944.00								8,400.00	5,080
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal

T.I.D Canal

Sample description: T.I.D Canal

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	215
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

June - 11/29/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	1,900.00	16,500.00		9.50
DL	100.00	100.00	100.00		0.01

June - 05/31/2023: Corn, silage

June - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	900.00	5,000.00		6.10
DL	100.00	100.00	100.00		0.01

Rooster - 10/28/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	1,700.00	14,700.00		9.60
DL	100.00	100.00	100.00		0.01

Rooster - 04/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	1,000.00	5,400.00		5.90
DL	100.00	100.00	100.00		0.01

Rooster - 08/19/2023: Sudangrass, silage

Sudan Silage

Sample and source description: Sudan Silage
Sample date: 10/26/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 76.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,600.00	1,000.00	10,900.00		16.00
DL	100.00	100.00	100.00		0.01

Sudan Silage

Sample and source description: Sudan Silage
Sample date: 10/26/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 73.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,400.00	900.00	8,500.00		14.00
DL	100.00	100.00	100.00		0.01

Tell - 10/28/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 57.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	2,000.00	16,000.00		9.10
DL	100.00	100.00	100.00		0.01

Tell - 04/21/2023: Corn, silage

Tell - 04/21/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	1,100.00	5,500.00		6.40
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	1,200.00	6,400.00		8.00
DL	100.00	100.00	100.00		0.01

Tell - 08/19/2023: Sudangrass, silage

Sudan Silage

Sample and source description: Sudan Silage

Sample date: 10/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,200.00	1,400.00	11,700.00		18.70
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

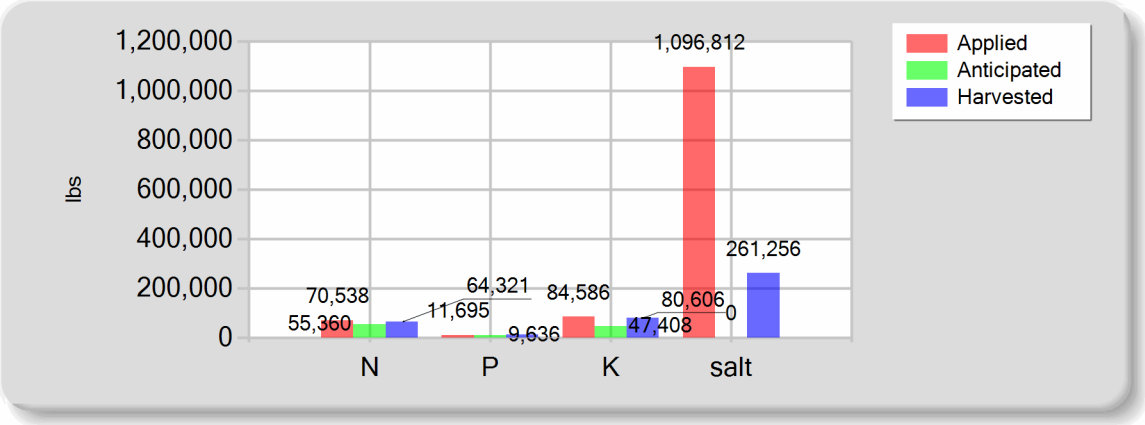
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

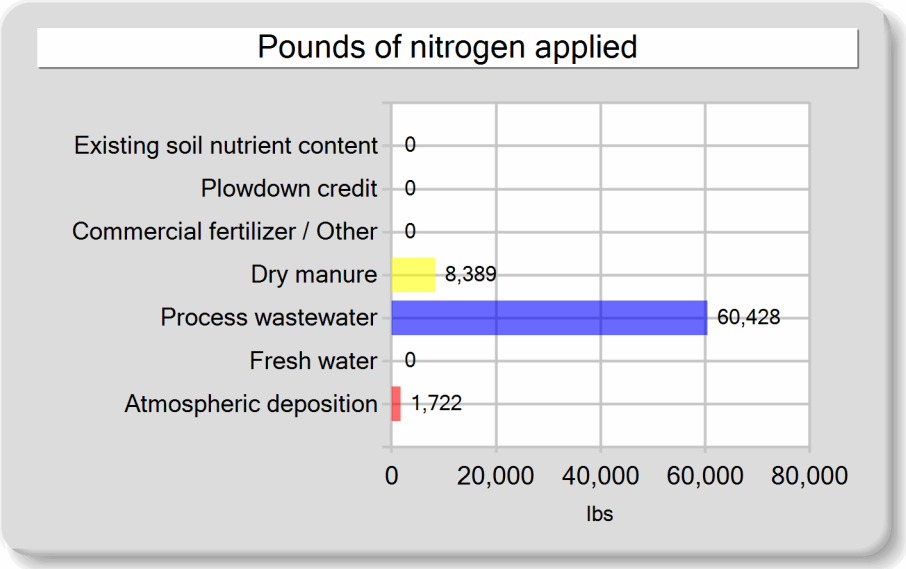
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

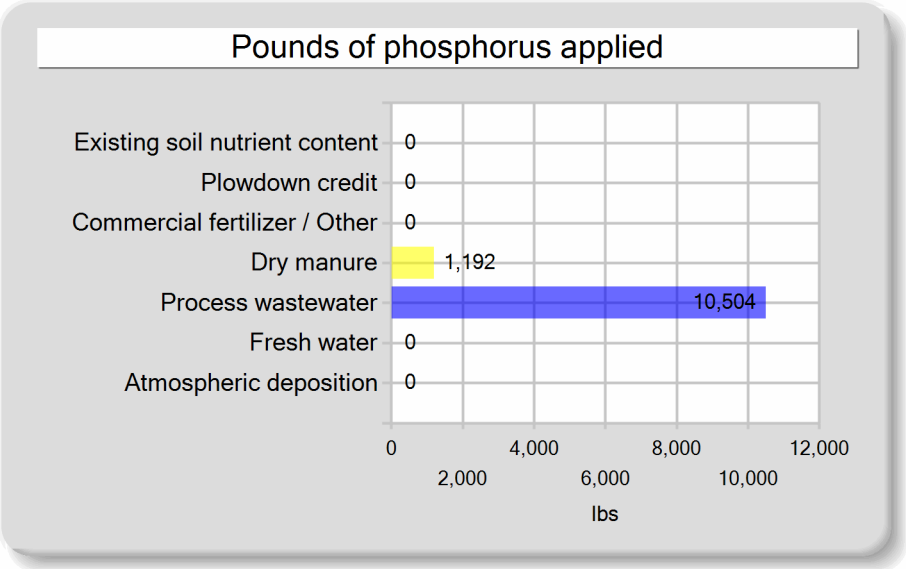
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	8,388.72	1,191.84	1,604.40	258,537.60
Process wastewater	60,427.62	10,503.56	82,982.10	518,502.68
Fresh water	0.00	0.00	0.00	319,771.43
Atmospheric deposition	1,722.00	0.00	0.00	0.00
Total nutrients applied	70,538.34	11,695.40	84,586.50	1,096,811.71
Anticipated crop nutrient removal	55,360.00	9,635.80	47,408.40	0.00
Actual crop nutrient removal	64,320.59	12,329.18	80,606.08	261,256.41
Nutrient balance	6,217.75	-633.78	3,980.41	835,555.30
Applied to removed ratio	1.10	0.95	1.05	4.20

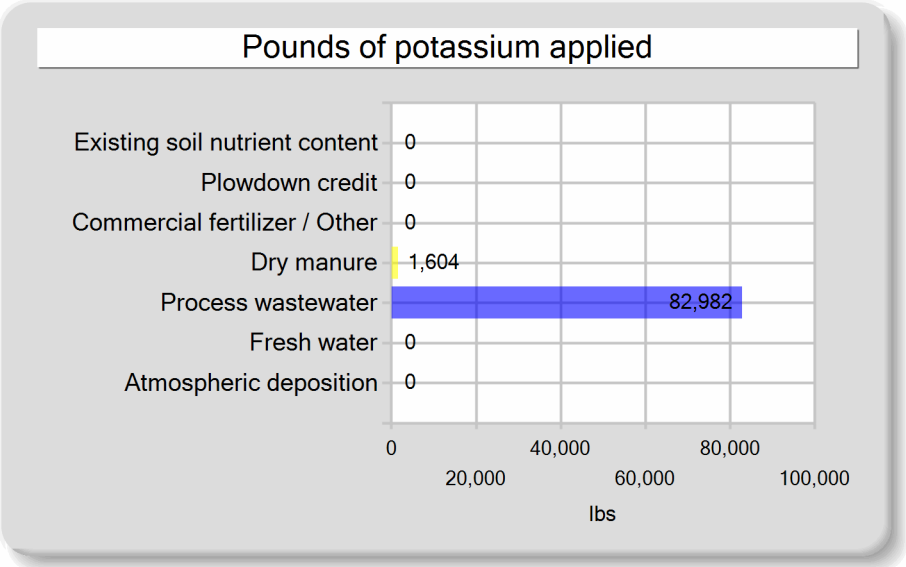
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

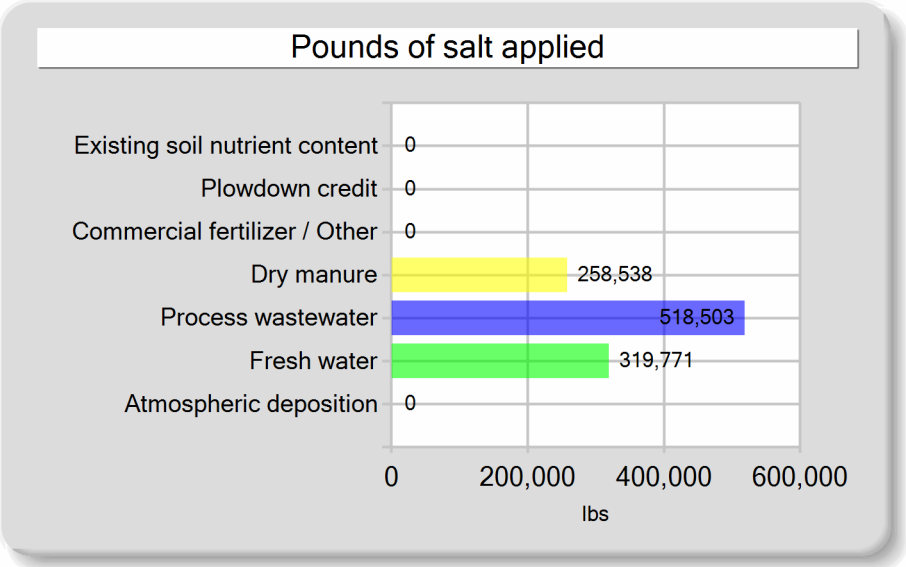


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Facility only houses beef animals onsite.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Kimberly Clauss Jorritsma Kimberly Clauss Jorritsma (May 28, 2024 14:41 PDT)	Kimberly Clauss Jorritsma Kimberly Clauss Jorritsma (May 28, 2024 14:41 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Kimberly Clauss-Jorristma	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 28, 2024	May 28, 2024
DATE	DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Clauss Dairy Farms
21670 Bloss Ave
Hilmar CA 95324
00-0022161 08

E-mail: omar@claussdairyfarms.com
Copy To: kathleen@claussdairyfarms.com;labs@fragservices.com

Lab No.: 23L0635
Sampled By: Omar H
Requested By: Omar Hernandez
Submitted Date: 12/12/23
Reported Date: 12/19/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	CDF Home	12/11/23	15:10	1290	1.29	8.8	*		7.4
2	CDF Clauss Last	12/11/23	15:15	1090	1.09	0.4	*		7.6
3	CDF Hosp. Barn	12/11/23	15:20	1220	1.22	0.1	*		7.5
4	CDF Milk Barn	12/11/23	15:25	1700	1.70	35.5	*		7.2
5	Valentine #1	12/11/23	15:30	2030	2.03	71.1	*		6.9
6	Valentine #2	12/11/23	15:25	2350	2.35	85.4	*		7.0
7	CDF Irrigation	12/12/23	8:30	1040	1.04	23.6	*		7.6

* = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Clauss Dairy Farms
21670 Bloss Ave
Hilmar CA 95324
00-0022161 08

E-mail: omar@claussdairyfarms.com
Copy To: kathleen@claussdairyfarms.com;labs@fragservices.com

Lab No.: 23L0635
Sampled By: Omar H
Requested By: Omar Hernandez
Submitted Date: 12/12/23
Reported Date: 12/19/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	CDF Home	12/11/23	15:10	**1290**	**1.29**	**8.8**	24.02	2.00
2	CDF Clauss Last	12/11/23	15:15	**1090**	**1.09**	0.4	1.09	0.09
3	CDF Hosp. Barn	12/11/23	15:20	**1220**	**1.22**	0.1	0.27	0.02
4	CDF Milk Barn	12/11/23	15:25	**1700**	**1.70**	**35.5**	96.92	8.08
5	Valentine #1	12/11/23	15:30	**2030**	**2.03**	**71.1**	194.10	16.18
6	Valentine #2	12/11/23	15:25	**2350**	**2.35**	**85.4**	233.14	19.43
7	CDF Irrigation	12/12/23	8:30	**1040**	**1.04**	**23.6**	64.43	5.37

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

2023 Annual Report- Valentine Dairy

Final Audit Report 2024-05-28

Created:	2024-05-22
By:	Kristen Jones (kjones@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAyMP7yRBhmcCIiQrf842NdYR7RRAHESN5

"2023 Annual Report- Valentine Dairy" History

Document created by Kristen Jones (kjones@fragservices.com)
2024-05-22 - 4:38:00 PM GMT

Document emailed to claussjerz@yahoo.com for signature
2024-05-22 - 4:38:50 PM GMT

Email viewed by claussjerz@yahoo.com
2024-05-22 - 4:39:26 PM GMT

Signer claussjerz@yahoo.com entered name at signing as Kimberly Clauss Jorritsma
2024-05-28 - 9:41:50 PM GMT

Document e-signed by Kimberly Clauss Jorritsma (claussjerz@yahoo.com)
Signature Date: 2024-05-28 - 9:41:52 PM GMT - Time Source: server

Agreement completed.
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